IMPACT - Modelling and optimisation of endless composite filament wound armour plates

Paul T. Kühner, Michal Habera, Andreas Zilian

University of Luxembourg

Submission type: Poster

Presenter: Paul T. Kühner (University of Luxembourg)

Sparse, wound carbon-reinforced structures are interesting for their high stiffness to weight ratio. Especially in the design of lightweight vehicle armour plates. However, the plates are often subject to a ballistic impact, which brings the structure close to (or beyond) its operational limits. This is contributed to by the onset of failure mechanisms such as buckling, fracture, shock-waves or other dynamic effects. We present a numerical model and optimisation of these structures. The implementation of the model developed during this project is open source and builds on top of `FEniCSx`. In particular employing non-standard functionality for the parallel ready implementation, including vertex integrals and branching manifolds. A novel geometry description of geometrically-discontinuous truss-like structures is used [1]. This description is in particular useful in the optimisation, due to a smaller number of intrinsic degrees-of-freedom.

References

1. Kühner, P. T., Habera, M., Zilian, A. (18 March 2026). Modelling of sparse lightweight composite structures using a geometrically discontinuous approach. GAMM 2026, Stuttgart, Germany.